

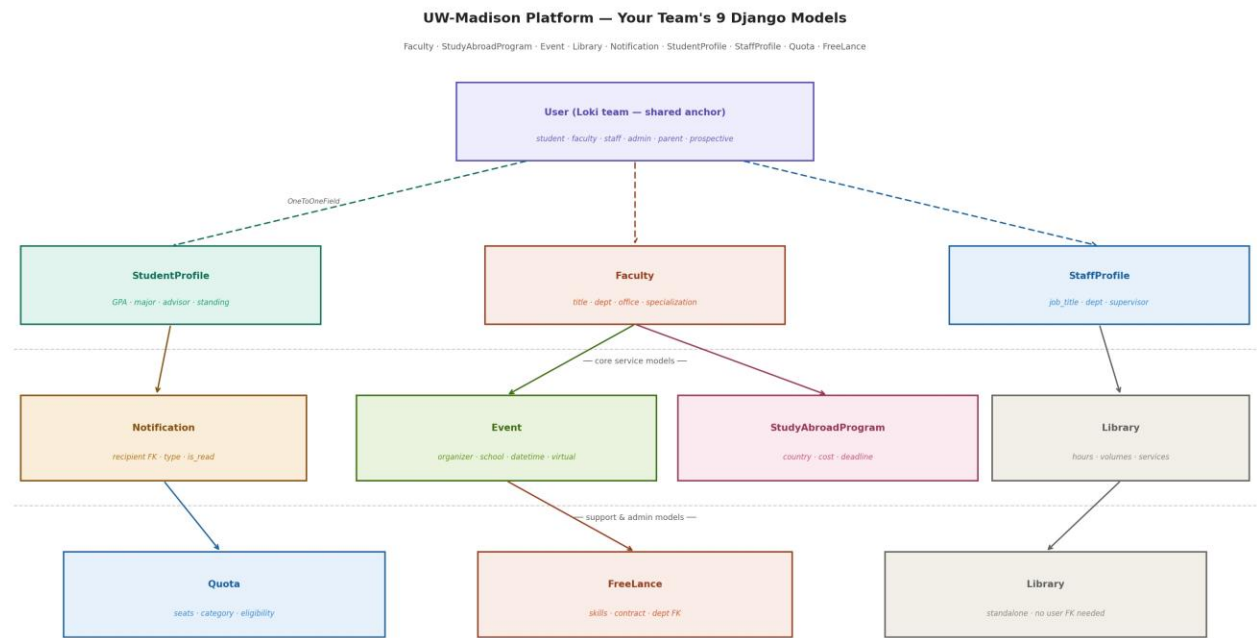
UW-Madison Our Team Explanation

Contents

- Section 1 — Team model flowchart (all 9 models + relationships)
- Section 2 — Faculty model
- Section 3 — StudentProfile model
- Section 4 — StaffProfile model
- Section 5 — Notification model
- Section 6 — Event model
- Section 7 — StudyAbroadProgram model
- Section 8 — Library model
- Section 9 — Quota model
- Section 10 — FreeLance model

Section 1 — Team Model Flowchart

The diagram below shows all 9 models your team owns, how they relate to each other, and how they connect to the shared User model (owned by Loki team). Dashed arrows = OneToOneField. Solid arrows = ForeignKey.



Relationship legend:

Model	Relationship type	How to use in code
StudentProfile	OneToOneField (User → Profile)	
Faculty	OneToOneField → User	Linked on DELETE CASCADE. Access via request.user.faculty
StaffProfile	OneToOneField → User	Linked on DELETE CASCADE. Access via request.user.staffprofile
Notification	ForeignKey → User (recipient)	Filter by recipient=request.user to get all alerts
Event	ForeignKey → User (organizer)	Nullable. filter(organizer=request.user) for own events
Freelance	ForeignKey → User (nullable)	Contractor may or may not have a platform login
Quota	GenericForeignKey → Course/Program/Housing	Uses ContentType — applies seat limits across models
StudyAbroad	Standalone — no User FK	All users browse programs; applications via other team
Library	Standalone — no User FK	Location/hours data shown on Student Dashboard

Section 2 — Faculty Model

What other models depend on Faculty?

- CourseSection — assigns an instructor to each class section
- StudentProfile — advisor FK points here
- Grade model — graded_by FK points here
- ResearchOpportunity — lead faculty researcher (other team)

Key fields explained

Field name	Django field type	Constraint	What it does & why
faculty_id	AutoField (PK)	PRIMARY KEY	Auto-generated unique ID for each faculty record.
user	OneToOneField(User)	ON DELETE CASCADE	Links to the User account. Delete the User and the Faculty profile deletes too.
employee_id	CharField(50)	UNIQUE	University HR employee number for payroll and admin systems.
title	CharField(50)	choices	Rank: Professor / Associate Professor / Assistant Professor / Lecturer. Shown on course pages and directories.
department	ForeignKey(Department)	nullable FK	The primary department this faculty member belongs to (Bela team model).
specialization	CharField(300)	nullable	Research or teaching focus. Example: Machine Learning. Shown on faculty profile pages.
office_location	CharField(200)	nullable	Building and room. Example: CS 4251. Students use this to find their professor.
office_hours	CharField(300)	nullable	When they are available. Example: Mon 2-4pm, Wed 10-12pm.
hire_date	DateField()	nullable	Date they joined UW. Used for seniority and tenure tracking.

Field name	Django field type	Constraint	What it does & why
is_active	BooleanField()	default=True	False = no longer employed. Historical course data is preserved.

Code example

```
faculty = Faculty.objects.get(user=request.user)
dept_faculty = Faculty.objects.filter(department_id=5, is_active=True)
```

Section 3 — StudentProfile Model

What features depend on StudentProfile?

- Student Dashboard — GPA, standing, advisor, credits shown at top
- Course registration — checks enrollment_status and academic_standing before allowing signup
- Advising portal — advisor FK lets faculty see their assigned students
- Financial aid — scholarship_info and is_international trigger eligibility checks
- Transcript model — links back to StudentProfile for each semester record

Key fields explained

Field name	Django field type	Constraint	What it does & why
student_id	AutoField (PK)	PRIMARY KEY	Unique ID for each student profile record.
user	OneToOneField(User)	ON DELETE CASCADE	Connects to the User account. One student = one User = one StudentProfile.
student_number	CharField(20)	UNIQUE, NOT NULL	Official university student ID shown on ID cards. Example: 9075432.
enrollment_status	CharField(30)	choices, NOT NULL	Full-time / Part-time / Withdrawn / Graduated. Controls access to services and financial aid.
academic_standing	CharField(30)	choices	Good Standing / Probation / Suspension / Honors. Determines whether student can register next semester.
class_level	CharField(20)	choices	Freshman / Sophomore / Junior / Senior / Graduate. Used for course eligibility checks.
gpa_cumulative	DecimalField(4,3)	nullable	Overall GPA from all completed semesters. Range 0.000 to 4.000.
gpa_semester	DecimalField(4,3)	nullable	GPA for the current semester only. Recalculated at semester close.

Field name	Django field type	Constraint	What it does & why
major	ForeignKey(Program)	nullable FK	Primary declared major. Points to the Program model (other team).
advisor	ForeignKey(Faculty)	nullable FK	The assigned academic advisor — points to your Faculty model.
expected_graduation	DateField()	nullable	Planned graduation date. Used by registrar for degree audits.
is_international	BooleanField()	default=False	True = student is on an international visa. Triggers compliance checks.
visa_type	CharField(30)	nullable	F-1, J-1, H-4 etc. Only filled when is_international = True.
scholarship_info	JSONField()	default=dict	Scholarships held. Example: {"name": "Hilldale", "amount": 3000, "year": 2025}.

Code example

```

profile = StudentProfile.objects.get(user=request.user)
if profile.academic_standing == "Probation":
    block_registration(profile)
print(profile.advisor.user.get_full_name())

```

Section 4 — StaffProfile Model

What other models depend on StaffProfile?

- VIP model — relationship_manager FK points to StaffProfile
- VVIP model — protocol_officer FK points to StaffProfile
- SSNRecord model — created_by FK points to StaffProfile
- StaffProfile itself — supervisor is self-referential FK for org charts

Key fields explained

Field name	Django field type	Constraint	What it does & why
staff_id	AutoField (PK)	PRIMARY KEY	Unique ID for each staff profile.
user	OneToOneField(User)	ON DELETE CASCADE	Connects to the User account. One staff = one User = one StaffProfile.
employee_id	CharField(50)	UNIQUE, NOT NULL	HR employee number. Separate from Faculty employee_id.
department	ForeignKey(Department)	nullable FK	The department this staff member is assigned to.
job_title	CharField(150)	NOT NULL	Official title. Example: Academic Advisor, IT Support Specialist, Facilities Manager.
employment_type	CharField(30)	choices	Full-time / Part-time / Contract / Temporary. Affects benefits and access rights.
office_location	CharField(200)	nullable	Building and room. Students and faculty use this to locate the staff member.
supervisor	ForeignKey(StaffProfile)	self-ref FK	Points to another StaffProfile — the reporting manager. Enables org chart queries.
hire_date	DateField()	nullable	Employment start date for anniversary and contract tracking.

Field name	Django field type	Constraint	What it does & why
contract_end_date	DateField()	nullable	Null for permanent staff. Set for contract/temporary workers.
is_active	BooleanField()	default=True	False = staff member has left. Data preserved for auditing.

Code example

```
advisors = StaffProfile.objects.filter(job_title__icontains="Advisor", is_active=True)
team = StaffProfile.objects.filter(supervisor=my_staff_profile)
```

Section 5 — Notification Model

What does Notification power?

- The notification bell / inbox on every dashboard (student, faculty, staff, admin)
- Email and SMS dispatch — reads notification_pref on User to pick the channel
- Grade posting alerts sent automatically when Faculty submit grades
- Enrollment change alerts when a student is added or dropped from a section
- Admin broadcast announcements pushed to all users at once

Key fields explained

Field name	Django field type	Constraint	What it does & why
notification_id	AutoField (PK)	PRIMARY KEY	Unique ID for each notification record.
recipient	ForeignKey(User)	ON DELETE CASCADE	The user who receives this alert. Deleting the User removes their notifications too.
title	CharField(200)	NOT NULL	Short headline shown in the bell. Example: Grade posted for CS 540.
message	TextField()	NOT NULL	Full notification body. Shown when user opens the notification in their inbox.
notification_type	CharField(50)	choices	Enrollment / Grade / Event / Alert / Announcement. Used to colour-code and filter the inbox.
is_read	BooleanField()	default=False	False = unread (shown bold). True = user has seen it. Toggle on click.
created_at	DateTimeField()	auto_now_add	When the notification was created. Used for sorting newest-first.
read_at	DateTimeField()	nullable	Set when user marks it read. Null = still unread.
action_url	URLField(500)	nullable	Deep link. Clicking the notification opens this URL. Example: /grades/course/540/.

Code example

```
Notification.objects.create(recipient=student_user, title="Grade posted for CS 540",  
    message="Your final grade has been submitted.", notification_type="Grade",  
    action_url="/grades/course/540/")  
unread = Notification.objects.filter(recipient=request.user, is_read=False).count()
```

Section 6 — Event Model

What features use Event?

- Student dashboard — Upcoming Events feed sorted by start_datetime
- VIP model — event_invitations field lists event IDs for exclusive invitations
- VVIP model — private_event_access lists restricted event IDs
- Notification model — event reminders triggered from Event records automatically

Key fields explained

Field name	Django field type	Constraint	What it does & why
event_id	AutoField (PK)	PRIMARY KEY	Unique ID for each event.
title	CharField(300)	NOT NULL	Event name shown in listings. Example: Spring Career Fair 2025.
event_type	CharField(50)	choices	Academic / Sports / Cultural / Career / Research / Social. Used for category filtering.
organizer	ForeignKey(User)	nullable FK	Admin or faculty who created the event. Nullable for system-generated events.
school	ForeignKey(School)	nullable FK	Associated school. Example: School of Engineering hosts an info session.
start_datetime	DateTimeField()	NOT NULL	When the event starts. Combined date + time for calendar integration.
end_datetime	DateTimeField()	NOT NULL	When it ends. Used to calculate duration and display on calendar grid.
location_name	CharField(200)	nullable	Venue. Example: Grainger Hall Auditorium.
is_virtual	BooleanField()	default=False	True = online event. When True, virtual_link is provided instead of location.
virtual_link	URLField(500)	nullable	Zoom / Teams / Webcast URL. Only shown to registered attendees if required.

Field name	Django field type	Constraint	What it does & why
max_attendees	IntegerField()	nullable	Seat limit. Null = unlimited. Registration closes automatically when full.
registration_required	BooleanField()	default=False	True = user must register. Enables attendance tracking and capacity control.
is_public	BooleanField()	default=True	False = restricted (VIP-only). Hidden from general browse unless user has access.

Code example

```
from django.utils import timezone
events = Event.objects.filter(start_datetime__gte=timezone.now(),
is_public=True).order_by("start_datetime")
my_events = Event.objects.filter(organizer=request.user)
```

Section 7 — StudyAbroadProgram Model

What features use StudyAbroadProgram?

- Study Abroad portal — students browse and filter by country, semester, cost, credits
- Admissions Application model — students apply to a specific program ID
- Admin panel — coordinators update program details and deadlines each term

Key fields explained

Field name	Django field type	Constraint	What it does & why
program_id	AutoField (PK)	PRIMARY KEY	Unique ID for each study abroad program.
name	CharField(300)	NOT NULL	Full program name. Example: UW in Tokyo — Language and Culture.
country	CharField(100)	NOT NULL	Host country. Primary filter students use when searching. Example: Japan.
city	CharField(100)	nullable	Host city. More specific than country. Example: Tokyo.
partner_institution	CharField(200)	nullable	Partner university. Important for credit transfer agreements. Example: Waseda University.
description	TextField()	nullable	Full program description — curriculum, housing, cultural immersion details.
duration_weeks	IntegerField()	nullable	Length in weeks. Example: 16 for a full semester.
semester	CharField(20)	choices	Fall / Spring / Summer. Determines which academic term the program runs.
credits_range	CharField(50)	nullable	Credit hours available. Example: 12-18 credits. Students need this for enrollment planning.
cost_estimate	DecimalField(10,2)	nullable	Estimated total cost in USD including tuition and housing. Key factor in student decisions.

Field name	Django field type	Constraint	What it does & why
application_deadline	DateField()	nullable	Last date to apply. After this date the program closes for applications.
max_participants	IntegerField()	nullable	Maximum students accepted. Null = no cap. Used for competitive programs.
is_active	BooleanField()	default=True	False = no longer offered. Hidden from browse but data preserved.

Code example

```

programs = StudyAbroadProgram.objects.filter(semester="Summer",
is_active=True).order_by("cost_estimate")
japan = StudyAbroadProgram.objects.filter(country="Japan", is_active=True)

```

Section 8 — Library Model

What features use Library?

- Student Dashboard — "Find a Library" widget showing hours and services
- Study room booking feature — has_study_rooms flag enables this per branch
- Library search — students filter by printing, study rooms, weekend hours

Key fields explained

Field name	Django field type	Constraint	What it does & why
library_id	AutoField (PK)	PRIMARY KEY	Unique ID for each library location.
name	CharField(200)	NOT NULL	Library name. Example: Memorial Library, STEM Library.
location_address	CharField(300)	nullable	Full address or building name. Used for map pins and walking directions.
contact_phone	CharField(20)	nullable	Main desk phone. Shown when students need to call ahead.
contact_email	EmailField(254)	nullable	Library email for room booking and enquiries.
website_url	URLField(500)	nullable	Official library site with catalogue and online resources.
hours_weekday	CharField(100)	nullable	Weekday hours. Example: Mon-Fri 7:30am-10pm. Shown on hours widget.
hours_weekend	CharField(100)	nullable	Weekend hours. Usually shorter. Example: Sat-Sun 10am-6pm.
total_volumes	IntegerField()	nullable	Total physical books and journals. Used for library profile statistics.
has_study_rooms	BooleanField()	default=False	True = students can book individual or group study rooms at this branch.

Field name	Django field type	Constraint	What it does & why
has_printing	BooleanField()	default=True	True = printing and copying available. Most libraries have this enabled.
is_active	BooleanField()	default=True	False = temporarily or permanently closed. Hidden from listings.
managed_by	ForeignKey(StaffProfile)	nullable FK	The librarian in charge of this branch. Points to StaffProfile. Null = no assigned manager yet.

Code example

```
libraries = Library.objects.filter(is_active=True, has_study_rooms=True).order_by("name")
weekend_libs = Library.objects.exclude(hours_weekend="").filter(is_active=True)
```

Section 9 — Quota Model

What problems does Quota solve?

- Reserve 20 seats in CS 540 specifically for international students
- Set a GPA minimum of 3.5 to qualify for a scholarship seat in a program
- Limit a housing unit to Graduate students only
- Auto-waitlist students when the quota is full instead of hard-rejecting them

Key fields explained

Field name	Django field type	Constraint	What it does & why
quota_id	AutoField (PK)	PRIMARY KEY	Unique ID for each quota rule.
quota_name	CharField(200)	NOT NULL	Descriptive name. Example: International Seats — CS 540 Fall 2025.
quota_type	CharField(50)	choices	Course / Program / Housing / Department — what kind of thing this quota applies to.
target_model	CharField(50)	choices	Model name the quota points at: Course, Program, HousingUnit, or Department.
target_id	IntegerField()	NOT NULL	Primary key of the specific object. Combined with target_model to identify the exact Course or Program.
category	CharField(50)	choices	General / International / Transfer / Graduate / Scholarship — which group this quota applies to.
total_seats	IntegerField()	NOT NULL	Total seats allocated under this quota rule.
reserved_seats	IntegerField()	default=0	How many seats are reserved for the specified category. Must be <= total_seats.
occupied_seats	IntegerField()	default=0	Currently filled seats. Updated on each enrollment action.
waitlisted_count	IntegerField()	default=0	Students waiting when quota is full. Used when auto_waitlist = True.

Field name	Django field type	Constraint	What it does & why
min_gpa_required	DecimalField(4,3)	nullable	Minimum GPA to use a reserved seat. Null = no GPA requirement.
auto_waitlist	BooleanField()	default=True	True = overflow students go to waitlist automatically rather than being rejected.
open_date	DateTimeField()	nullable	When registration opens for this quota.
close_date	DateTimeField()	nullable	When this quota closes — no more enrollments after this date.

Code example

```

from django.contrib.contenttypes.models import ContentType
course = Course.objects.get(code="CS 540")
ct = ContentType.objects.get_for_model(Course)
quota = Quota.objects.get(content_type=ct, object_id=course.id, category="International")
if quota.occupied_seats < quota.reserved_seats:
    quota.occupied_seats += 1
    quota.save()

```

Section 10 — FreeLance Model

Why does the university need FreeLance?

- Central record of who is contracted to which department and for what project
- Contract start and end dates with deliverable deadline tracking
- Tax compliance — tax_id enables Finance to generate IRS Form 1099 for payments over \$600
- Department heads see which contractors are currently active and their ratings

Key fields explained

Field name	Django field type	Constraint	What it does & why
freelance_id	AutoField (PK)	PRIMARY KEY	Unique ID for each contractor record.
user	ForeignKey(User)	nullable FK	Optional platform login. Some contractors have accounts, others do not.
full_name	CharField(200)	NOT NULL	Contractor name — required even without a User account.
email	EmailField(254)	UNIQUE	Contact email. Unique to prevent duplicate contractor entries.
expertise_area	CharField(300)	NOT NULL	Main service. Example: Web Development, Data Analysis, Graphic Design.
skills	JSONField()	default=list	Skill list. Example: ["Python", "React", "PostgreSQL"]. Used to match contractors to projects.
portfolio_url	URLField(500)	nullable	Portfolio or LinkedIn link. Used during contractor vetting.
contract_type	CharField(30)	choices	Hourly / Fixed / Project-based / Retainer. Determines billing and deliverable structure.
hourly_rate	DecimalField(8,2)	nullable	USD rate per hour. Set when contract_type = Hourly.
project_rate	DecimalField(10,2)	nullable	Fixed total price. Set when contract_type = Fixed.

Field name	Django field type	Constraint	What it does & why
department	ForeignKey(Department)	nullable FK	Which department hired this contractor. Charges costs to the right budget.
deliverables	JSONField()	default=list	Expected outputs with deadlines. Example: [{"item": "Redesign homepage", "due": "2025-06-01"}].
tax_id	CharField(50)	nullable	Tax ID for IRS Form 1099. Required by Finance for payments over \$600.
is_active	BooleanField()	default=True	True = contract running. False = completed or terminated.
rating	DecimalField(3,2)	nullable	Quality score 0.00 to 5.00. Set by department after project completion.

Code example

```
from datetime import date, timedelta
contractors = FreeLance.objects.filter(department_id=7,
is_active=True).order_by("contract_end_date")
expiring = FreeLance.objects.filter(contract_end_date__lte=date.today()+timedelta(days=30),
is_active=True)
```